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10 Mbit Ethernet protocol: From RTL to routed GDS

A beginner's walkthrough of a checked open-source reference build

Follow ethernet_core from its communication rules and Verilog **source** to a routed IHP SG13G2 layout. Worked calculations, measured digital traces and tool results are identified separately. This is an educational block, not a manufactured chip or a signoff-ready design.

STAGE	WHAT IT PRODUCED	RESULT
Simulation	Listed directed checks	PASS directed
Synthesis	1042 mapped functional cells	PASS mapping
Routing	Routed DEF and extracted SPEF	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	Decoded placement agreement	PASS export
Foundry DRC / LVS	No accepted result	UNPERFORMED

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What is in this document

Start with the wire-level rules, then follow the same circuit through the tools. Each section has a specific evidence boundary.

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SECTION 1 · START HERE

Start here

A protocol is an agreement about how connected devices exchange information: which wire carries each bit, when to sample it, and how to recognise a complete transfer.

WHAT WE BUILT AND WHAT IT PROVES

A Verilog description represents circuits, not software instructions running one after another. Registers remember values; combinational logic calculates the next values. A **clock** advances the registered state. The next pages first explain this protocol at its wires, then connect those rules to the actual reference RTL.

The implemented unit is ethernet_core in hw/protocols/stretch/ethernet. It is a fixed **reference** block, not the Jane Street programmable combined emulator or a Tiny Tapeout submission top. No new competition architecture, instruction set, pin map or authority decision is implied by these documents.

No external PHY, MDIO, collision handling or network stack. Test clocks are phase-offset at the same nominal frequency; drift is untested. Later pages give the exact stimulus and the implementation evidence rather than using a broad protocol-compliant label. A successful mapped netlist, routed DEF and decoded **GDS** establish concrete implementation/export results, not readiness to manufacture.

The reading path starts with chip-design vocabulary, then develops the protocol, calculations, local interface and failure cases. Separate sections explain lint, simulation, synthesis, placement, routing, timing and **GDS**. The last two pages define design and physical terms. No prior ASIC-flow experience is assumed.

EVIDENCE VOCABULARY

Label	Meaning in this document
Worked example	Explicit assumptions; independently calculated
Measured trace	Positive digital regression VCD, where present
Directed PASS	Listed assertions and negative controls
Routed / decoded GDS	Physical export and structural agreement
Signoff	Not performed or claimed

SECTION 2 · CHIP DESIGN

Chip design in ten minutes

Each stage makes the same circuit more concrete. The folders below are the actual unit layout, not a copied APB command list.

CHIP DESIGN

RTL states what hardware should do. A simulator checks selected input sequences before any silicon exists. Synthesis replaces that description with gates from a process library. Physical tools choose positions and metal routes. Timing analysis checks the resulting delays; stream-out writes the mask-layout geometry.

THE PIPELINE, STAGE BY STAGE

FOLDER	WHAT HAPPENS HERE
00_lint	Read code for risky patterns; retained warnings
01_rtl	Pinned source and the local wrapper
02_sim	Independent stimulus, comparisons and fatal controls
03_verification	Supplemental checks; not a universal UVM PASS
04_synthesis	Map functional logic to SG13G2 cells
05_dft	Manufacturing test work remains unperformed
06_floorplanning	Die outline, rows, power and pin locations
07_placement	Choose legal positions for mapped cells
09_place_and_route	Clock tree, routing and SPEF extraction
08_sta	Read post-route TT and slow timing reports
10_gds	Stream-out and independent placement decoder
11_physical_verification	Foundry DRC / LVS remain unperformed

SECTION 3 · INTERFACE BOUNDARY

A MAC is not a 10BASE-T PHY

The block implements full-duplex digital MII frame processing.

INTERFACE BOUNDARY

MII transfers four bits per clock. At 2.5 MHz the 400 ns clock period carries one nibble, giving 10 Mbit/s. Bytes cross the interface low nibble first, then high nibble. For byte A5 the visible nibbles are 5 followed by A. This ordering differs from simply reading a printed hexadecimal byte left to right.

A real link still needs an external analogue PHY, suitable magnetics and connector circuitry. Manchester line coding, link negotiation, MDIO management and half-duplex collision/retry are not supplied by this digital **reference**. Ethernet addressing and FCS processing do not automatically provide an IP, TCP or UDP stack.

The wrapper contains separate TX and RX clocks. TX data and handshake belong to mii_tx_clk; received output belongs to mii_rx_clk. The consumer cannot assume that simultaneous-looking frames share a safe common register domain.

MII OWNERSHIP

Signal group	Clock / owner
mii_txd, tx_en, tx_er	TX clock; MAC drives
mii_rxd, rx_dv, rx_er	RX clock; external PHY drives
tx_data / ready-valid	TX clock local producer
rx_data / valid / last / user	RX clock local consumer

SECTION 4 · FRAME FORMAT

Preamble, padding and FCS

The test checks frame structure separately from the local payload stream.

FRAME FORMAT

The transmitter emits the synchronising preamble and SFD before the supplied frame bytes. With padding enabled and a minimum 64-byte frame length including FCS, an input shorter than sixty frame bytes is extended before the four-byte FCS is appended. The local input bytes already contain whatever destination/**source**/type fields the application supplies.

A fourteen-byte input therefore needs forty-six padding bytes to reach sixty, then four FCS bytes. A sixty-byte input needs no padding. A 256-byte input exceeds the minimum and retains its full body. The regression checks those three sizes and independently evaluates their transmitted CRC values.

At 10 Mbit/s, the minimum 64-byte frame itself takes 51.2 us. Including eight preamble/SFD bytes and a twelve-byte inter-frame gap gives 84 byte-times, or 67.2 us of wire scheduling. These are calculated line-rate bounds, not measured application throughput or a **payload** efficiency claim.

WORKED SIZE BUDGET

Input frame bytes	Padding	FCS	Result
14	46	4	64 bytes
60	0	4	64 bytes
256	0	4	260 bytes

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SECTION 5 · CRC CALCULATION

A reflected eight-step transform

The integrated replacement updates the exact state output consumed by these two MAC engines.

CRC CALCULATION

The byte update first XORs the eight data bits into the low end of the 32-bit state. Each of eight steps shifts right and conditionally XORs EDB88320 if the departing low bit is one. For the usual Ethernet convention the initial state is FFFFFFFF and the final result is complemented. ASCII 123456789 then gives **CRC32** CBF43926.

The earlier generic LFSR elaboration was expensive. The replacement narrows the implementation to this consumed byte/state contract. Its state_out is compared against the preserved upstream LFSR; its unused data_out is tied to zero. It is therefore not a drop-in generic LFSR replacement for other consumers.

This mapping is a linear transformation of forty input bits: thirty-two state bits and eight data bits. Zero plus all forty one-hot basis vectors establish agreement for a truly linear implementation; 256 deterministic mixed vectors add a useful diagnostic cross-check. Fixed nonzero CRC initialisation and final complement occur around, not inside, that linear transform.

SOURCE / COMMANDS

```
assign stage[0]=state_in^{24'b0,data_in};
genvar b;
generate for(b=0;b<8;b=b+1)begin: crc_bit
assign stage[b+1]=(stage[b]>>1)^(stage[b][0]?32'hedb88320:32'b0);
end endgenerate
assign state_out=stage[8];
```

01_rtl/ethernet_crc32.v:18; exact contiguous source excerpt

SECTION 6 · RTL INTERFACE

Stream input, non-stallable receive output

The same-looking byte ports have different buffering obligations.

RTL INTERFACE

On TX, the producer supplies data, valid, last and user and respects ready. These signals belong to the transmit **clock**. The wrapper fixes MII mode, padding enabled, minimum frame length 64 and a twelve-byte gap. Error-underflow remains available, but a passing normal-frame test is not an underflow regression.

On RX, data, valid, last and user belong to the receive **clock**. There is no rx_ready input, so downstream must accept each valid output or provide a suitable buffer in the same domain. Crossing to a system clock requires an explicit CDC design such as a correctly sized asynchronous FIFO, not a casual combinational connection.

Reset is synchronous active high in both domains. The integration layer must arrange safe assertion and release relative to each clock. The reference constraints apply reset budgets to both clock definitions; those constraints do not by themselves establish system reset reliability. The current SDC defines both MII clocks without set_clock_groups -asynchronous. It therefore retains a synchronous clock relationship in STA rather than modelling arbitrary external PHY phase and frequency. The common reset input has delays in both domains, but safe asynchronous reset delivery/release is not proved by those budgets.

SECTION 7 · CONTROL SEQUENCE

Two domains and phase-offset tests

Independent receive stimuli avoid a self-confirming TX-to-RX loop.

CONTROL SEQUENCE

The bench runs TX with a 400 ns period and RX with the same nominal 400 ns period after an initial 73 ns offset. Their phases differ; their frequencies do not drift. This establishes separate stimulus scheduling and exercises the distinct MAC domains, but it is not a frequency-offset or asynchronous-FIFO stress test.

The receive **source** creates its own frames and CRC values rather than wiring transmitted nibbles back to the DUT. Valid sixty- and 256-byte receive frames are checked. A corrupt-FCS frame must raise the expected receive error. The transmitter independently checks fourteen-, sixty- and 256-byte input frames, padding and FCS.

The control path must keep last aligned with the final supplied frame byte. The FCS and pad extension are generated by the MAC. End-of-frame on **MII** and end-of-input on the local stream therefore occur at different times.

DOMAIN-SPECIFIC TRANSACTIONS

Path	Independent directed cases
TX	14, 60 and 256 input frame bytes
RX	60 and 256 valid frame bytes
RX fault	Corrupt FCS
Clock relation	400 ns each; 73 ns phase offset

SECTION 8 · FAILURE ANALYSIS

CRC success does not cover every fault

The untested conditions define the next regression work.

FAILURE ANALYSIS

A corrupted receive FCS is covered, but RXER, runt/oversize acceptance, transmit underflow and **clock** drift are not established. Half-duplex collisions cannot be claimed because this reference is full duplex and does not implement collision/retry logic. Physical link negotiation and board operation likewise belong to the external PHY integration.

A consumer that cannot drain non-stallable RX output may lose data even when the incoming FCS is valid. A production integration needs a buffer-capacity budget based on maximum burst size and consumer latency. Declaring two clocks in STA is not a substitute for proving the cross-domain queue contract.

The fixed CRC replacement has a deliberately limited compatibility promise. Preserving its unused data_out zero tie is acceptable for the checked consumers, but reusing the module under a different generic-LFSR interface would change the contract and require a new review.

NEXT REGRESSION OBLIGATIONS

Area	Unclosed condition
Receive faults	RXER, runt / oversize
Transmit flow	Underflow and error injection
Clock crossing	Frequency drift and buffer stress
PHY / board	MDIO, negotiation, actual link
Half duplex	Not implemented

SECTION 9 · WORKED EXAMPLE

Byte A5 becomes nibbles 5, A

This is a labelled ideal MII sequence, not an exported waveform measurement.

WORKED EXAMPLE

At a 2.5 MHz **MII** clock, each nibble occupies 400 ns and one byte occupies 800 ns. The byte A5 appears as data nibble 5 then nibble A while transmit-enable is asserted. A known preamble/SFD sequence is checked before the frame body so a correctly ordered payload is not accepted in an incorrectly framed transmission.

The normal-frame tests compute FCS independently from the supplied frame and any generated zero padding. The low-nibble-first **MII** ordering must also hold for the FCS bytes. A local byte comparison alone would miss that serialisation error, which is why the bench observes the nibble interface.

The current Ethernet bench retains decisive logs and **source** rather than a plotted positive VCD. The document therefore shows the explicit example as a table. A future board trace must identify PHY clock direction, enable timing and error lines before being described as a measured link result.

IDEAL TIMING FOR BYTE A5

MII cycle	Relative interval	Nibble
First	0..400 ns	5
Second	400..800 ns	A
Recovered byte	After two cycles	A5

SECTION 10 · LINT

Lint: read code before running it

Verilator checks legal but suspicious code patterns without exercising transactions. A warning count is a review starting point, not proof of correctness.

LINT

The retained report contains 72 warning messages and 0 error messages. Categories present: CASEINCOMPLETE, PINCONNECTEMPTY, PINMISSING, PROCASSINIT, SELRANGE, UNUSEDPARAM, UNUSEDSIGNAL, WIDTHEXPAND, WIDTHTRUNC. These figures count report message headers; repeated warnings are not necessarily independent defects. The report is hashed in this document evidence manifest.

The invocation deliberately uses -Wno-fatal so warnings remain visible while lint execution can complete. A successful command therefore does not mean every warning was resolved. Width changes, incomplete cases, **reset** use and unconnected ports need review against the wrapper contract and directed tests.

Lint cannot show that a receive byte is correct, that a transaction finishes, or that an external electrical interface is safe. Those questions require a checking testbench, implementation evidence and separate board or signoff work.

RETAINED WARNING REVIEW

MEASURE	VALUE
Warning message headers	72
Error message headers	0
Clean warning-free RTL?	Not claimed
Protocol behaviour proved by lint?	No

SOURCE / COMMANDS

```
make lint
# inspect, do not just count
# 00_lint/lint_report.txt
```

00_lint/Makefile and run_flow.sh: retained Verilator report

SECTION 11 · SIMULATION AND VERIFICATION

Independent frame and CRC checks

A testbench is code that drives the circuit and judges its outputs. These directed checks use independent expected data and deliberately failing controls.

SIMULATION AND VERIFICATION

Directed frame tests assert preamble, SFD, low-nibble-first ordering, padding, **payload** and independent CRC32 for TX sizes fourteen, sixty and 256. RX uses independently generated sixty- and 256-byte frames and a corrupt-FCS case. Reset quiescence is checked, and a deliberately wrong expected receive byte must fail.

The dedicated CRC regression compares the consumed state_out against the preserved upstream LFSR over zero, forty one-hot and 256 mixed vectors, checks the standard arithmetic anchor and rejects an injected mismatch. This addresses the local optimisation separately from frame-level behaviour.

The result remains a full-duplex digital MAC **reference**. It is not a complete Ethernet device, a network stack, a frequency-drift regression or foundry signoff. Test coverage is documented per condition, so future work can add RXER and underflow without pretending the current normal-frame run already covers them.

ACCEPTED PROOF LAYERS

Layer	Evidence
Frame behaviour	Independent TX / RX checks
CRC transform	297 direct comparisons plus anchor
Checker sensitivity	Deliberate fatal mismatches
Not established	All faults, PHY operation, signoff

> **A checker must be able to fail**
>
> Completion alone is insufficient. Read the successful log and the expected fatal rejection together; neither is a protocol-compliance certificate.

SECTION 12 · SYNTHESIS

Synthesis: a description becomes gates

A netlist is a list of logic cells and their connections. Yosys translates the RTL into cells from the IHP SG13G2 library.

SYNTHESIS

A process design kit, or PDK, contains the process-specific files used by implementation tools. This **reference** uses IHP SG13G2, not the SKY130 library used by the golden APB primer. Standard cells are predesigned gates and storage elements with characterised timing and physical views.

The **source** is compiled and checked under the unit workspace before synthesis. Yosys maps the functional logic into the pinned IHP SG13G2 standard-cell library. The current netlist contains 1042 functional cells, excluding $scopeinfo metadata. The structural checker rejects remaining generic logic, latches or unresolved instances.

The mapped count excludes $scopeinfo metadata. It is not the count of every placed instance after **clock** buffers, repair cells and fillers are added. It also does not mean that the original protocol has been independently proven for all possible transactions.

MAPPING RESULT

MEASURE	CURRENT RESULT
Top module	ethernet_core
Mapped functional cells	1042
Technology	IHP SG13G2
Unresolved generic logic / latches	Rejected by structural checker

SOURCE / COMMANDS

```
make synth
node verify_build.mjs synth
```

04_synthesis/synth.tcl; verify_build.mjs synthesis checks

SECTION 13 · PHYSICAL IMPLEMENTATION

Floorplan and placement

The floorplan gives the circuit a physical outline. Placement chooses where its cells sit inside that outline.

PHYSICAL IMPLEMENTATION

The die outline for this standalone block is 500 x 500 um. The floorplan establishes rows and **reference** power/pin geometry. The placer must fit cells legally without overlaps and with room for routing. This is a block-level educational layout, not a complete padframe or package design.

Mapped functional logic has 1042 cells. After physical additions and fillers, the routed DEF contains 53,628 placed standard-cell instances. A filler occupies empty row sites; it is not another functional logic gate. That distinction explains why the two counts differ so much.

Utilisation is the fraction of core area occupied by placed cell area. It is not simply functional-cell count divided by die size, and it changes between stages. This note does not substitute APB utilisation, row count or I/O count for unavailable protocol-specific measurements.

ACTUAL PHYSICAL REFERENCE

MEASURE	VALUE
Die outline	500 x 500 um
Mapped functional cells	1042
Placed standard-cell instances	53,628
Physical inputs	06_floorplanning; 07_placement
Placement agreement	DEF compared with decoded GDS

SECTION 14 · CLOCK AND WIRES

Clock tree and routing

Clock-tree synthesis distributes clocks to registers. Routing draws the metal connections between placed cells.

CLOCK AND WIRES

A register samples its input at a **clock** edge. Clock buffers distribute that edge across the block; the difference in arrival times is called skew. The reference is constrained at MII TX/RX: 2.5 MHz / 400 ns. TX and RX have separate MII clock definitions; their relationship is a reference assumption, not arbitrary external PHY CDC signoff.

OpenROAD completes the staged physical flow. The **reference** die is 500 x 500 um. DEF records routed design geometry and cell placement; SPEF captures extracted parasitics used by the timing analysis. The scripts retain stage-specific output so the result can be investigated instead of relying on a final banner.

Real interconnect has resistance and capacitance, called parasitics. Extraction writes these in **SPEF** for post-route timing. The retained router-rule DRC file is empty. That checks the router view of its work; it is not a full foundry-deck check of the exported polygons.

READ THE PHYSICAL EVIDENCE

ARTIFACT	QUESTION IT ANSWERS
09_place_and_route/route.def	Where are cells and routed connections?
09_place_and_route/route.spef	Which parasitics were extracted?
Router-rule DRC report	Any reported routing-rule violations?
08_sta/timing_report.txt	What does extracted reference timing report?

> **Two checks can both be called DRC**
>
> The empty router report is accepted routing evidence. Foundry DRC is a separate unperformed obligation; one result cannot stand in for the other.

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SECTION 15 · REFERENCE TIMING

Two measured reference corners

Static timing analysis adds cell and wire delays without simulating input patterns. Setup asks whether data arrives early enough; hold asks whether it stays stable long enough.

REFERENCE TIMING

The physical **clock** constraint is MII TX/RX: 2.5 MHz / 400 ns. Post-route timing uses the extracted design rather than an ideal pre-route wire estimate. The report at each corner must be inspected for worst setup and hold slack and for VIOLATED markers. The reported values are not a timing guarantee under unstated board, voltage or temperature conditions.

Current TT setup/hold slack is +396.77 / +0.18 ns. The slow-corner values are +395.00 / +0.27 ns. Both reports contain no VIOLATED marker. These numbers are taken directly from the local report files at document build time; the evidence manifest retains their hashes.

I/O delays are **reference** budgets, not a contract with an external target, transceiver or PHY. Bus synchronisation, reset recovery/removal, all additional corners and modes, signal integrity and board timing need their own analysis. Positive slack in these two reports is therefore called reference timing closure, not foundry or system signoff. For Ethernet specifically, the SDC does not declare the two MII domains asynchronous. The quoted slacks use that reference relationship and do not bound arbitrary inter-domain paths or reset behaviour for independent real PHY clocks. No asynchronous CDC or reset timing signoff is claimed.

LIVE EXTRACTED SLACK, NS

Corner	Worst setup	Worst hold
TT	396.77	0.18
Slow	395.00	0.27
Status	PASS reference reports	No VIOLATED markers

> **Positive slack has a stated boundary**
>
> Slack is remaining timing margin. These values apply to the listed clocks, I/O budgets and two library corners; they are not a board-timing or manufacturing signoff guarantee.

SECTION 16 · GDS STREAM-OUT

Count the right objects

GDS stores layout polygons and hierarchical cell references. Stream-out combines routed geometry with the real standard-cell layouts from the PDK.

GDS STREAM-OUT

The **GDS** file contains 3,219,696 bytes and 47 structures. The independent decoder finds 53,628 placed standard-cell instances including fillers. That placement count is intentionally much larger than the 1042 mapped functional cells: filler instances occupy unused placement sites and structures are unique hierarchy definitions.

The decoder compares exported cell identities, coordinates and orientations against the routed DEF. A deliberately altered placement/export expectation fails, including under Python optimisation, so acceptance is not implemented with removable assert-only checks. Raw file SHA256 is a run-specific fingerprint because stream-out timestamps can change.

The OpenROAD router-rule DRC report is empty. It is not a foundry-deck DRC result. LVS, antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing signoff remain unperformed. No empty routing report is promoted into a DRC-clean tapeout claim.

LIVE EXPORT MEASUREMENTS

Quantity	Current value
Mapped functional cells	1042
GDS bytes	3,219,696
GDS structures	47
Placed standard-cell instances	53,628
Die	500 x 500 um

SECTION 17 · UNPERFORMED SIGNOFF

DRC and LVS: what is still missing

Design rule checking asks whether geometry follows the process rules. Layout versus schematic asks whether the extracted layout circuit matches the intended netlist.

UNPERFORMED SIGNOFF

Foundry DRC requires the applicable, qualified rule deck and an inspected report on the final polygons. Typical checks include minimum widths, spacing and enclosure. No foundry-deck DRC result has been accepted for this protocol **reference**. The router report is narrower evidence.

LVS requires extracting a circuit from the layout and comparing devices and connectivity with the intended circuit. A structural **GDS** decoder checking cell references, coordinates and orientations does not perform that extraction or connectivity comparison. No accepted LVS match is available.

Antenna, scan/DFT, power integrity, padframe, tap/endcap strategy and manufacturing release remain unperformed. The golden APB primer reports its own separate checks; those results do not transfer to SG13G2 or to this design.

SEPARATE ACCEPTED EXPORT FROM MISSING SIGNOFF

CHECK	THIS REFERENCE
Router-rule report	Empty; narrower routing evidence
Decoded GDS placement	PASS structural/export agreement
Foundry-deck DRC	UNPERFORMED
Extracted-layout LVS	UNPERFORMED
Manufacturing readiness	Not claimed

> **A GDS file is not tapeout approval**
>
> This package is suitable for studying and reproducing a block implementation. Do not hand it off as a signoff-clean manufacturing design.

SECTION 18 · RESULTS SUMMARY

Everything at a glance

Read each result together with the question it actually answers. The same PASS label never implies every protocol or manufacturing obligation is closed.

RESULTS SUMMARY

The directed functional result and the physical export result are both useful, but they prove different things. The physical build uses the fixed wrapper configuration and **reference** constraints. Additional modes, external peripherals, arbitrary clock relationships and board operation require their own evidence.

Important protocol boundary: No external PHY, MDIO, collision handling or network stack. Test clocks are phase-offset at the same nominal frequency; drift is untested.

FULL REFERENCE FLOW

STAGE	TOOL / EVIDENCE	STATUS
Lint	Verilator; 72 retained warnings	Review
Simulation	Independent directed expectations	PASS directed
Negative controls	Deliberate fatal mismatches	PASS rejection
Synthesis	Yosys; 1042 mapped cells	PASS mapping
Floorplan / placement	OpenROAD; 500 x 500 um	PASS reference
Routing / extraction	DEF and SPEF retained	PASS reference
Timing	TT and slow setup / hold	PASS reference
GDS	KLayout; 3,219,696 bytes	PASS export
Foundry DRC / LVS	No accepted report	UNPERFORMED
Full-chip / board	No accepted integration proof	UNPERFORMED

SECTION 19 · RUN IT YOURSELF

Run it yourself in WSL

Use the pinned source, the unit tool lock and the external PDK revision together.

RUN IT YOURSELF

The proven entry point is a WSL login Bash shell. igny init prepares the packaging runtime. Unsetting CRUCIBLE_ENV after initialisation allows the unit crucible.toml workspace selection to control tool resolution rather than an inherited environment override. This is the observed workflow, not a change to Crucible internals.

Each unit has its existing environment and lock. The common tool set includes Icarus, Verilator, Yosys, OpenROAD and klayout-stream. The separate PDK checkout is IHP SG13G2 revision 22f43352dd8219f9007eb659e422e0d5fe28c5fb. A tool lock does not pin that external PDK by itself; IHP_PDK_ROOT must point to the matching installation.

The documented machine could not use the pip klayout package as the expected stream-out executable; igny tool install klayout-stream supplies the proven converter. make lint sim checks functional behaviour, make flow executes the staged implementation, and node verify_build.mjs all inspects artifacts. Review normal and intentionally failing logs before accepting the result.

Re-running a simulation can replace its VCD and invalidate an earlier document trace hash. Retain a document evidence manifest or regenerate the measured figures after the positive run. Do not silently preserve an old chart while claiming it is the newly generated waveform.

SOURCE / COMMANDS

```
igny init
unset CRUCIBLE_ENV
cd hw/protocols/stretch/ethernet
igny workspace sync
igny tool install klayout-stream
make lint sim
make flow
node verify_build.mjs all
```

Run after entering the repository in WSL login Bash; PDK required.

SECTION 20 · SOURCES AND REMAINING WORK

Sources, ownership and remaining work

The pinned revision identifies the baseline; local integration changes are described separately.

SOURCES AND REMAINING WORK

Upstream: https://github.com/alexforencich/verilog-ethernet
Pinned commit: 77320a9471d19c7dd383914bc049e02d9f4f1ffb
Licence: MIT, Alex Forencich; per-file notices retained. Original **source** notices remain in the RTL. The document is authored by Ignytion; this does not transfer ownership of upstream code or remove its licence obligations.

Local modifications, directed test stimulus and **source** boundaries are recorded in hw/protocols/stretch/ethernet/README.md and the repository append-only .context.md. The first three protocol lessons have separate editable 16:9 student decks; this A4 technical note is a reading and evidence document, not a replacement for their instructor notes.

Primary reading: The checked-in KSZ8081MLX PHY data sheet for **MII**, IEEE 802.3 as the canonical standard, and the pinned MAC RTL.
Canonical standard catalogue: https://standards.ieee.org/standard/802_3-2022.html; local PHY snapshot: specs/Ethernet_Specification.pdf

Open work remains explicit: No external PHY, MDIO, collision handling or network stack. Test clocks are phase-offset at the same nominal frequency; drift is untested. Full-chip integration, manufacturing checks and board validation are not closed by the standalone **reference** flow. The reader can reproduce the listed evidence without assuming those larger obligations have already been completed.

SECTION 21 · DESIGN VOCABULARY

Glossary: design and protocol

Definitions for reading the protocol rules and the checked RTL. Physical implementation terms follow on the next page.

DESIGN AND VERIFICATION

TERM	WHAT IT MEANS
Protocol	Rules for exchanging information between devices
RTL	Registers and the logic moving values between them
Verilog	A language used to describe hardware
FSM	A circuit progressing through defined states
Lint	Static source checks; no transaction execution
Testbench	Code that supplies inputs and checks outputs
Coverage	Which defined situations were exercised
Negative control	An intentionally wrong case the checker must reject
VCD	Digital signal-change trace from a simulation
Handshake	Signals agreeing when data is accepted
MAC / PHY	Digital frame engine / physical signalling transceiver
MII	Four-bit Media Independent Interface
FCS	Frame Check Sequence; transmitted CRC32
IFG	Inter-frame gap
Basis vector	One active input bit used to test a linear transform

SECTION 22 · PHYSICAL VOCABULARY

Glossary: implementation and signoff

These definitions help distinguish a successfully exported reference layout from a manufacturing-ready design.

PHYSICAL IMPLEMENTATION

TERM	WHAT IT MEANS
Synthesis	Translate RTL into a technology-cell circuit
Netlist	Cells and the connections between them
Standard cell	Predesigned logic/storage cell in a process library
PDK	Process design kit: process-specific implementation files
SG13G2	IHP process used by these reference builds
Floorplan	Die/core outline, rows, power and pin geometry
Placement	Choose legal positions for cells
Filler	Physical row-site cell, not functional logic
CTS / skew	Clock distribution / difference in arrival times
Routing / via	Metal connections / links between metal layers
Parasitics / SPEF	Wire resistance/capacitance / extracted data format
Corner	A characterised process, voltage and temperature case
STA / slack	Static timing analysis / remaining timing margin
DEF / GDS	Placement/routing exchange / polygon hierarchy
DRC	Check geometry against applicable process rules
LVS	Compare the extracted layout circuit with its netlist
Signoff	Accepted applicable release checks; unperformed here

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